

Rohit Padwal

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Summary

Skilled software engineer with 6 years of experience designing, developing, and delivering backend and distributed systems across enterprise domains. Proficient in Java, Python, microservices, and cloud-native technologies, with a track record of improving system performance, reliability, and deployment efficiency. Experienced in full-stack development, CI/CD pipelines, and leading cross-functional teams to deliver production-grade software.

Work Experience

Software Engineer

IBM – Instana

KL, India

Jul 2026 – Present

- Designed and deployed an **incremental machine-learning retraining framework** on k8s that adapts to new multi-tenant log patterns (triggered at 10K+ unmatched logs, capped at 450K sampling), maximizing template coverage and anomaly-detection accuracy while reducing false alerts by 35%.
- Scaled a **high-throughput Java inference engine** integrated with a Python ML service, enabling real-time template matching and low-latency anomaly detection across multi-tenant log ingestion pipelines
- Generated **AI-assisted analysis** (Claude & Copilot) and optimization tools for testing and documentation.

Software Engineer

IBM – Instana

TX, USA

Jan 2024 – Jul 2026

- Implemented **OpenTelemetry integration** driven by Java backend services, improving log data accuracy, limiting troubleshooting time, and increasing overall distributed system reliability.
- Engineered a **scalable log pipeline** utilizing Kafka, gRPC, ClickHouse, and Dropwizard, and leveraged **microservices architecture** to speed up template iteration and troubleshooting, simplifying log processing latency by 10%.
- Developed **log ingestion solutions** using Kafka streaming and Elasticsearch storage for high-performance analytics.
- Built a Python-based AI/NLP log comprehension pipeline to extract structured templates automatically from millions of unstructured log lines per day, reducing log analysis effort by 60% & enabling scalable, tenant-aware pattern detection.

Software Engineer Intern

Ipser Labs LLC

CA, USA

Sep 2023 – Dec 2023

- Increased performance of Java & React applications, trimming page load time by 25% through enhancements.
- Refactored REST APIs and SQL queries using AI-driven recommendations and **automated code review**, amplifying response time and overall user experience.
- Streamlined and tested **EC2, S3, Lambda deployments**, elevated scalability, and cost optimization strategies.

Software Engineer Intern

RepairSmith by AutoNation

CA, USA

Jun 2022 – Aug 2022

- Researched and optimized **GitLab CI/CD pipelines** using Groovy scripts, refining CRM system performance by 10%.
- Practiced Agile methodologies within an 8-member team and applied **scalable architecture patterns and Docker-based containerization**, lowering deployment time by 30% and enhancing release consistency.
- Accelerated **API performance and backend efficiency** by refactoring service logic and optimizing database interactions, reducing response latency.

Senior Software Engineer

Infosys Limited – FedEx

MH, India

Dec 2019 – Jul 2021

- **Led a team** of 3 developers, managing **end-to-end SDLC** and delivering 2 critical projects within scope, while programming in Java (Spring Boot, microservices), Angular, and SQL to rectify & upgrade applications, resulting in a 20% gain in application stability & performance.
- Added **centralized logging** and metrics collection (Jaeger), improving observability across distributed services.

- Engineered deployment using **Docker**, **Kubernetes**, **AWS**, and Jenkins for CI/CD, stabilizing client's UI perfectly, cutting deployment ops time by 30%.
- Reorganized **OAuth** for secure authentication and authorization in development of web applications by Angular Material UI, Node, and Bootstrap over Windows and Linux, accomplishing a seamless and protected user experience, contributing to 30% reinforced security.
- Modernized Java Spring Boot microservices for **integration of Kafka for data exchange**, contributing to a 25% decrease in data transfer latency.

Software Engineer

Product Dossier Solutions India Pvt Ltd

MH, India

May 2018 – Jun 2019

- Executed Java-Spring, **REST**, & Angular for debugging, business logic, and troubleshooting for **Project Management ERP** Software as a full-stack developer in software development life cycle, yielding bug fixing, reliability, and overall performance of software by 20%.
- Utilized **J2EE** & SQL to develop ERP, achieving a 5% cut down in data processing & integration with external systems.
- Coded Spring Boot & **Hibernate** for building scalable applications, delivered 20% growth in database query performance.
- Modernized JSP pages with React & JavaScript to enrich user interface, leading to a 15% increase in user interaction.

Education

M.S. in Computer Science · *University of Texas at Arlington*

May 2023

GPA: 3.9 / 4.0 Coursework: Web Development, Cloud Computing, Software Engineering, Databases, Distributed Systems

B.E. in Computer Engineering · *Savitribai Phule Pune University*

July 2017

Coursework: Algorithm Design, Data Structures, Object-Oriented & Reactive Programming

Skills

Languages	Java, Python, PHP, Go, Bash
Databases	SQL, PostgreSQL, Oracle, MySQL, NoSQL, ClickHouse
Cloud	AWS, Microsoft Azure, GCP, IBM Cloud
Web & Frontend	Angular, React, HTML5, CSS3, JavaScript, JSP, Node.js, TypeScript, AJAX, jQuery
Tools & Frameworks	Jira, Git, Maven, Kafka, JUnit, Spring Boot, Docker, Kubernetes, Helm, Jaeger, OpenShift, REST API, Flask, OpenTelemetry, Tomcat, Mockito, Jenkins, Prometheus, Postman, SonarLint, Swagger, IntelliJ IDEA, SonarQube, SDLC, Hibernate, Gradle
AI / ML	LLM, NLP, Prompt Engineering, Agentic Workflow

Projects

- **SolGram: Grammar Fuzzer for Ethereum Smart Contracts** (Python, Angular, Solidity, GCP) — Developed a grammar-based fuzzer to discover vulnerabilities in Ethereum smart contracts.
- **Cryptographic Algorithm Analysis: Performance & Security Evaluation** (Python, Cryptography, Data Analysis) — Performance and security evaluation of multiple cryptographic algorithms with comparative benchmarking.
- **Sorting Algorithm GUI & Analysis: Interactive Visualization Tool** (Java/Python, Data Structures, GUI) — Interactive visualization tool for real-time comparison of sorting algorithm performance.
- **Desktop Voice Assistant: Voice-Controlled Automation Tool** (Python, Speech Recognition, NLP) — Voice-controlled automation tool enabling hands-free interaction with desktop applications.

Certifications

- [OTCA: OpenTelemetry Certified Associate](#)
- [IBM AI Developer Professional Certificate](#)